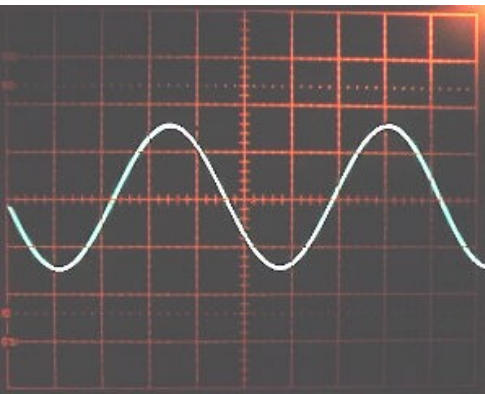


# ECE 3317

Dr. Jiefu Chen

## Notes 2

# Complex Vectors



Adapted from notes by Prof. Stuart A. Long

# Notation

- $V(t)$  is a time-varying function.
- $V$  is a phasor (complex number).
- A bar underneath indicates a vector:  $\underline{V}(t)$ ,  $\underline{V}$ .

Appendices [A](#), [B](#), [C](#), and [D](#) in the Shen & Kong text book list frequently used symbols and their units. (scanned and attached in the end of the slides)

# Complex Numbers

$$c = a + j b = |c| e^{j\phi}$$

$\uparrow$                        $\uparrow$                        $\uparrow$                        $\uparrow$   
 Real part      Imaginary part      Magnitude      Phase (always in radians)

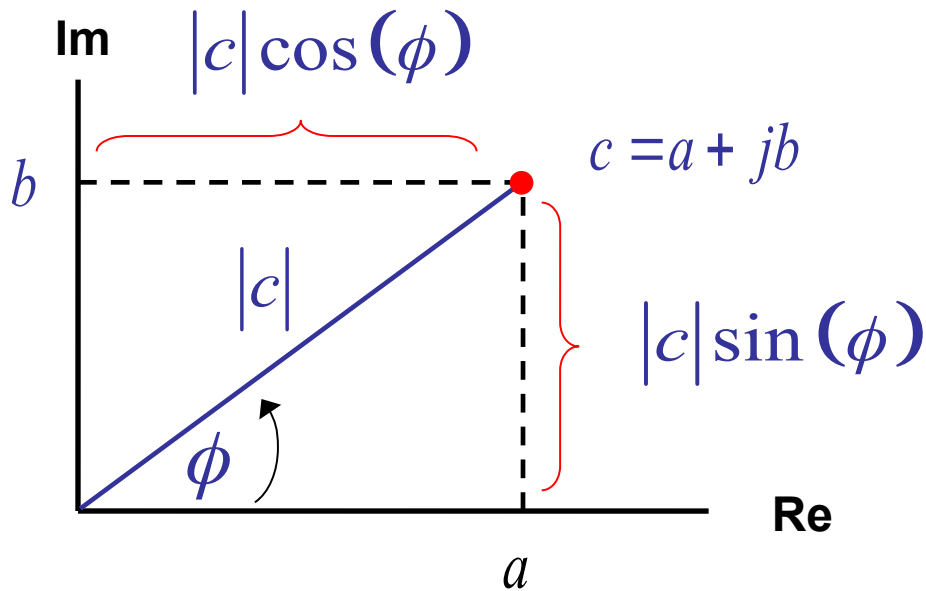
$j = \sqrt{-1}$

$$|c| = \sqrt{a^2 + b^2}$$



Euler's identity:

$$e^{j\phi} = \cos \phi + j \sin \phi$$



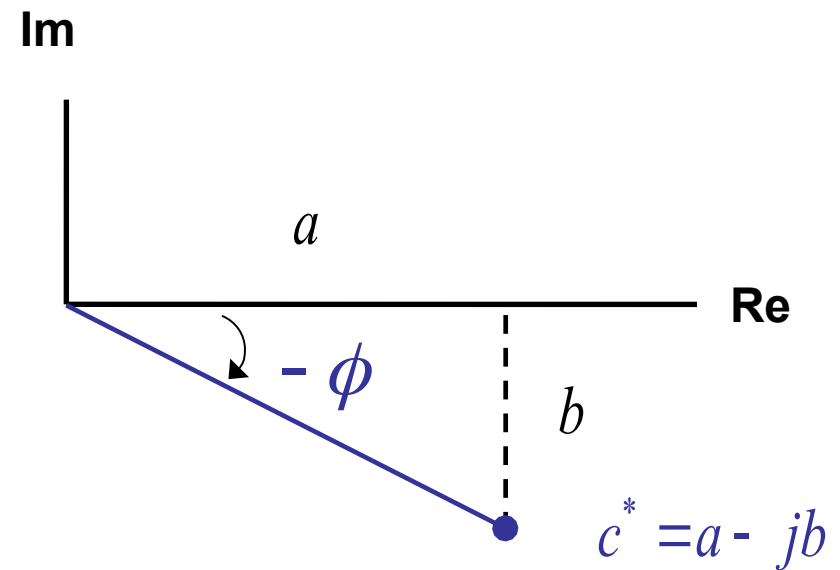
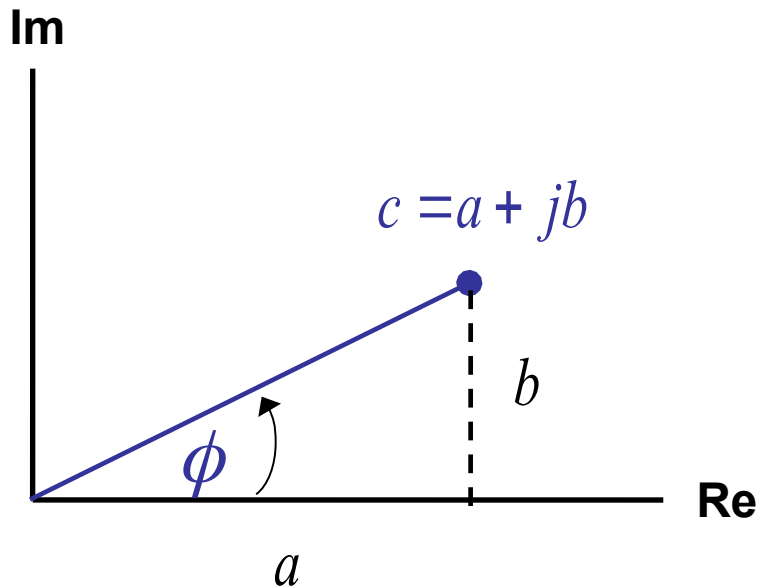
$$a = |c| \cos \phi$$

$$b = |c| \sin \phi$$

# Complex Numbers

## Complex conjugate

$$c^* = (a + jb)^* \equiv a - jb = |c| e^{-j\phi}$$



# Complex Algebra

$$c_1 = a_1 + jb_1 = |c_1| e^{j\phi_1}$$

$$c_2 = a_2 + jb_2 = |c_2| e^{j\phi_2}$$

## Addition

$$c_1 + c_2 \equiv (a_1 + a_2) + j(b_1 + b_2)$$

## Subtraction

$$c_1 - c_2 \equiv (a_1 - a_2) + j(b_1 - b_2)$$

## Complex Algebra (cont.)

$$\begin{aligned}c_1 &= a_1 + jb_1 = |c_1| e^{j\phi_1} \\c_2 &= a_2 + jb_2 = |c_2| e^{j\phi_2}\end{aligned}$$

### Multiplication

$$(c_1)(c_2) \equiv |c_1||c_2| e^{j(\phi_1 + \phi_2)}$$

### Division

$$\frac{c_1}{c_2} \equiv \frac{|c_1|}{|c_2|} e^{j(\phi_1 - \phi_2)}$$

# Square Root

$$c = |c| e^{j\phi}$$

Principle square root

$$\sqrt{c} = (|c| e^{j\phi})^{1/2}$$
$$\sqrt{|c|} e^{j(\phi/2)}$$

$$-\pi < \phi \leq \pi$$

(principal branch)

$$\begin{aligned} c^{1/2} &= (|c| e^{j\phi})^{1/2} \\ &= (|c| e^{j(\phi+2\pi n)})^{1/2} \\ &= \sqrt{|c|} e^{j(\phi/2 + n\pi)} \\ &= (\sqrt{|c|} e^{j\phi/2}) e^{jn\pi} \\ &= \pm \sqrt{c} \end{aligned}$$

where  $n$  is an integer

*The complex square root will have two possible values.  
(The principal branch is unique.)*

# Time-Harmonic Quantities

$$V(t) = V_0 \cos(\omega t + \phi)$$

Amplitude

Angular  
Frequency

Phase

$$f = \frac{\omega}{2\pi}$$

From Euler's identity:

$$V(t) = \operatorname{Re}\left\{ V_0 e^{j\phi} e^{j\omega t} \right\}$$

$$T = \frac{1}{f}$$

Define the phasor :  $V \equiv V_0 e^{j\phi}$

We then have

$$V(t) = \operatorname{Re}\left\{ V e^{j\omega t} \right\}$$

## Time-Harmonic Quantities (cont.)

$$V(t) = V_0 \cos(\omega t + \phi)$$

$$\left\{ \begin{array}{ll} \mathbf{V} = V_0 e^{j\phi} & \text{going from time domain to phasor domain} \\ V(t) = \operatorname{Re}\left\{ \mathbf{V} e^{j\omega t} \right\} & \text{going from phasor domain to time domain} \end{array} \right.$$

Time-domain  $\Leftrightarrow$  Phasor domain

$$V(t) \Leftrightarrow \mathbf{V}$$

## Time-Harmonic Quantities (cont.)

$$V(t) \Leftrightarrow \mathbf{V}$$

Note :

$$U(t) + V(t) \Leftrightarrow \mathbf{U} + \mathbf{V}$$

This assumes that the two sinusoidal signals are at the same frequency.

$$\frac{\partial}{\partial t} V(t) \Leftrightarrow j\omega \mathbf{V}$$

There are no time derivatives in the phasor domain!

However,

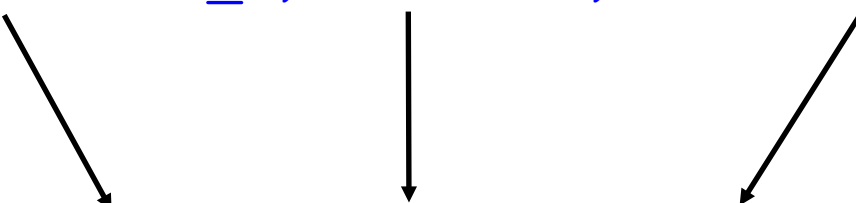
$$U(t)V(t) \not\Leftrightarrow \mathbf{U}\mathbf{V}$$

All phasors are complex numbers, but not all complex numbers are phasors!

# Complex Vectors

Transform each component of a time-harmonic vector function into a *complex vector*.

$$\underline{V}(t) = \underline{\hat{x}}V_x \cos(\omega t + \phi_x) + \underline{\hat{y}}V_y \cos(\omega t + \phi_y) + \underline{\hat{z}}V_z \cos(\omega t + \phi_z)$$


$$\underline{V} \equiv \underline{\hat{x}}V_x e^{j\phi_x} + \underline{\hat{y}}V_y e^{j\phi_y} + \underline{\hat{z}}V_z e^{j\phi_z}$$

To see this:

$$\begin{aligned}\underline{V}(t) &= \underline{\hat{x}}V_x \cos(\omega t + \phi_x) + \underline{\hat{y}}V_y \cos(\omega t + \phi_y) + \underline{\hat{z}}V_z \cos(\omega t + \phi_z) \\ &= \text{Re} \left\{ \left( \underline{\hat{x}}V_x e^{j\phi_x} + \underline{\hat{y}}V_y e^{j\phi_y} + \underline{\hat{z}}V_z e^{j\phi_z} \right) e^{j\omega t} \right\} \\ &= \text{Re} \left\{ \underline{V} e^{j\omega t} \right\}\end{aligned}$$

Hence

$$\underline{V}(t) \Leftrightarrow \underline{V}$$

## Example 1.15 (Shen & Kong)

Assume  $\underline{A} = \underline{\hat{x}} + j\underline{\hat{y}}$

Find the corresponding time-domain vector

$$|\underline{A}(t)| = \sqrt{A_x^2 + A_y^2} = \sqrt{\cos^2 \omega t + \sin^2 \omega t} = 1$$

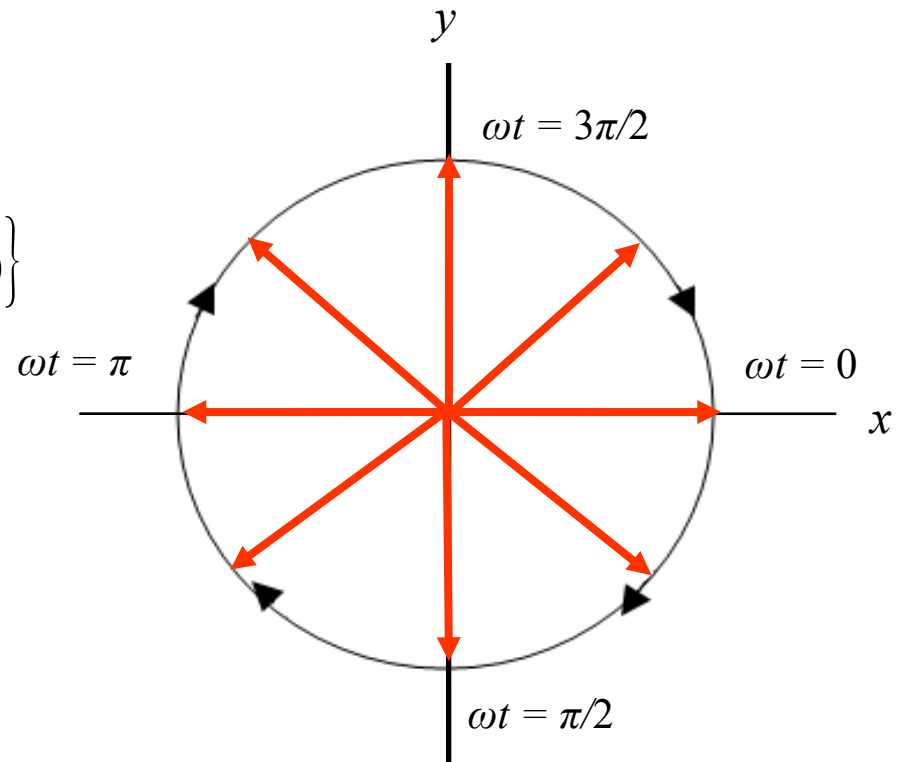
$$\underline{A}(t) = \text{Re} \left\{ \underline{A} e^{j\omega t} \right\}$$

$$\underline{A}(t) = \text{Re} \left\{ (\underline{\hat{x}} + j\underline{\hat{y}}) e^{j\omega t} \right\}$$

$$\underline{A}(t) = \text{Re} \left\{ (\underline{\hat{x}} + j\underline{\hat{y}})(\cos \omega t + j \sin \omega t) \right\}$$

$$\underline{A}(t) = \underline{\hat{x}} \cos \omega t - \underline{\hat{y}} \sin \omega t$$

$$\underline{A}(t) = \underline{\hat{x}} \cos \omega t - \underline{\hat{y}} \sin \omega t$$



The vector rotates with time!

# Time Average of Time-Harmonic Quantities

$$V(t) = V_0 \cos(\omega t + \phi)$$

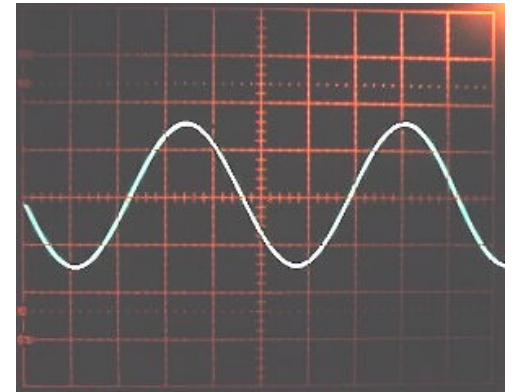
$$\langle V(t) \rangle \equiv \frac{1}{T} \int_0^T V_0 \cos(\omega t + \phi) dt = 0$$

$$\left( T = \frac{1}{f} = \frac{2\pi}{\omega} \right)$$

$$\langle V^2(t) \rangle = \frac{1}{T} \int_0^T V_0^2 \cos^2(\omega t + \phi) dt$$

$$\langle V^2(t) \rangle = \frac{V_0^2}{T} \int_0^T \left\{ \frac{1 + \cos[2(\omega t + \phi)]}{2} \right\} dt$$

$$\langle V^2(t) \rangle = \frac{V_0^2}{T} \left( \frac{T}{2} \right) = \frac{V_0^2}{2}$$



Hence

$$\langle V^2(t) \rangle = \frac{V_0^2}{2}$$

## Time Average of Time-Harmonic Quantities (cont.)

Next, consider the time average of a **product** of sinusoids:

$$\langle V(t)I(t) \rangle = \frac{1}{T} \int_0^T [V_0 \cos(\omega t + \alpha)] [I_0 \cos(\omega t + \beta)] dt$$

$$\begin{aligned} \langle V(t)I(t) \rangle &= \frac{1}{T} V_0 I_0 \int_0^T \cos(\omega t + \alpha) \cos(\omega t + \beta) dt \\ &= \frac{V_0 I_0}{T} \int_0^T \left[ \frac{\cos(\alpha - \beta) + \cos(2\omega t + \alpha + \beta)}{2} \right] dt \\ &= \frac{V_0 I_0}{T} \left[ T \frac{\cos(\alpha - \beta)}{2} \right] \\ &= V_0 I_0 \left[ \frac{\cos(\alpha - \beta)}{2} \right] \end{aligned}$$

↑  
Sinusoidal (time ave = 0)

# Time Average of Time-Harmonic Quantities (cont.)

Next, consider

$$\begin{aligned} \mathbf{VI}^* &= (V_0 e^{j\alpha}) (I_0 e^{j\beta})^* \\ &= (V_0 e^{j\alpha}) (I_0 e^{-j\beta}) \\ &= V_0 I_0 e^{j(\alpha - \beta)} \end{aligned}$$

Hence,

$$\operatorname{Re}(\mathbf{VI}^*) = V_0 I_0 \cos(\alpha - \beta)$$

Recall that

$$\langle V(t)I(t) \rangle = V_0 I_0 \left[ \frac{\cos(\alpha - \beta)}{2} \right] \quad (\text{from previous slide})$$

Hence

$$\langle V(t)I(t) \rangle = \frac{1}{2} \operatorname{Re}(\mathbf{VI}^*)$$

## Time Average of Time-Harmonic Quantities (cont.)

The results directly extend to **vectors** that vary sinusoidally in time.

Consider:  $\underline{D}(t) \cdot \underline{E}(t) = (D_x E_x + D_y E_y + D_z E_z)$

$$D_{x,y,z}(t) = \text{Re} \left[ D_{x,y,z} e^{j\omega t} \right] \text{ etc.}$$

$$\begin{aligned} \langle \underline{D} \cdot \underline{E} \rangle &= \langle D_x E_x \rangle + \langle D_y E_y \rangle + \langle D_z E_z \rangle \\ &= \frac{1}{2} \text{Re}(D_x E_x^*) + \frac{1}{2} \text{Re}(D_y E_y^*) + \frac{1}{2} \text{Re}(D_z E_z^*) \\ &= \frac{1}{2} \text{Re}(D_x E_x^* + D_y E_y^* + D_z E_z^*) \end{aligned}$$

Hence  $\langle \underline{D}(t) \cdot \underline{E}(t) \rangle = \frac{1}{2} \text{Re}(\underline{D} \cdot \underline{E}^*)$

## Time Average of Time-Harmonic Quantities (cont.)

The result holds for both **dot** product and **cross** products.

$$\langle \underline{D}(t) \cdot \underline{E}(t) \rangle = \frac{1}{2} \operatorname{Re}(\underline{D} \cdot \underline{E}^*)$$

$$\langle \underline{E}(t) \times \underline{H}(t) \rangle = \frac{1}{2} \operatorname{Re}(\underline{E} \times \underline{H}^*)$$

$$D(x, y, z, t) = \operatorname{Re} \left[ \underline{D}(x, y, z) e^{j\omega t} \right] \text{ etc.}$$

## Time Average of Time-Harmonic Quantities (cont.)

To illustrate, consider the **time-average stored electric energy** for a sinusoidal field.

$$U_E(t) = \frac{1}{2} \underline{D}(t) \cdot \underline{E}(t) \quad (\text{from ECE 3318})$$

$$\begin{aligned} \langle U_E(t) \rangle &= \frac{1}{2} \langle \underline{D}(t) \cdot \underline{E}(t) \rangle \\ &= \frac{1}{2} \left[ \frac{1}{2} \text{Re}(\underline{D} \cdot \underline{E}^*) \right] \end{aligned}$$

$$\langle U_E(t) \rangle = \frac{1}{4} \text{Re}(\underline{D} \cdot \underline{E}^*)$$

## Time Average of Time-Harmonic Quantities (cont.)

Similarly,

$$\langle U_H(t) \rangle = \frac{1}{4} \text{Re}(\underline{\underline{B}} \cdot \underline{\underline{H}}^*)$$